

MARS

EARL C. SLIPHER



THE PHOTOGRAPHIC STORY



A COLOR PHOTOGRAPH OF MARS produced by the well-known Kodak Dye Transfer Process from a tri-color set of black and white separation negatives taken with the 27½-inch telescope alternately through blue, yellow-green and red transmission filters. Original photographs were made on July 4, 1954, by the National Geographic Society-Lowell Observatory expedition to the Lamont-Hussey observatory, Bloemfontein, South Africa. (The repeated copying of this photograph in the reproduction processes has greatly increased the color contrast.)

FOREWORD

Less than two years after the short hop of the first airplane flight at Kitty Hawk, E. C. Slipher began making photographs of the planets at Lowell Observatory. Within the year of his most recent observations (1961), a space probe passed within 62,000 miles of Venus. As this book goes to press other probes are planned which are intended to pass even closer to Venus and Mars. Between these epoch-making events the author has carried on a continuous visual and photographic study of the brighter planets.

In accordance with a tradition established by the work of Percival Lowell, the founder of the Lowell Observatory, Slipher has devoted most of this effort to the study of the "Red Planet".

The value of his unique photographic record of Mars stems from his ceaseless effort and meticulous care. One must not think, however, that the results are based on these qualities alone. When his highly trained eye perceived the unusual, he made every effort to obtain a permanent photographic record and to follow the event throughout its life history. Although much is unavoidably lost in reproduction, the quality of the plates contained in this book attest to his photographic skill. Slipher was the first to make photographs systematically in different spectral regions and the first to discover the advantages of superimposing a number of images obtained under nearly identical conditions. Above all, he was the pioneer who recognized many important changes on Mars.

Many of his photographs have been published in textbooks and scientific journals and many more have been given to anyone who requested them. Although the great majority of the photographs presented here were obtained by the author, he has included some of the best obtained by other investigators.

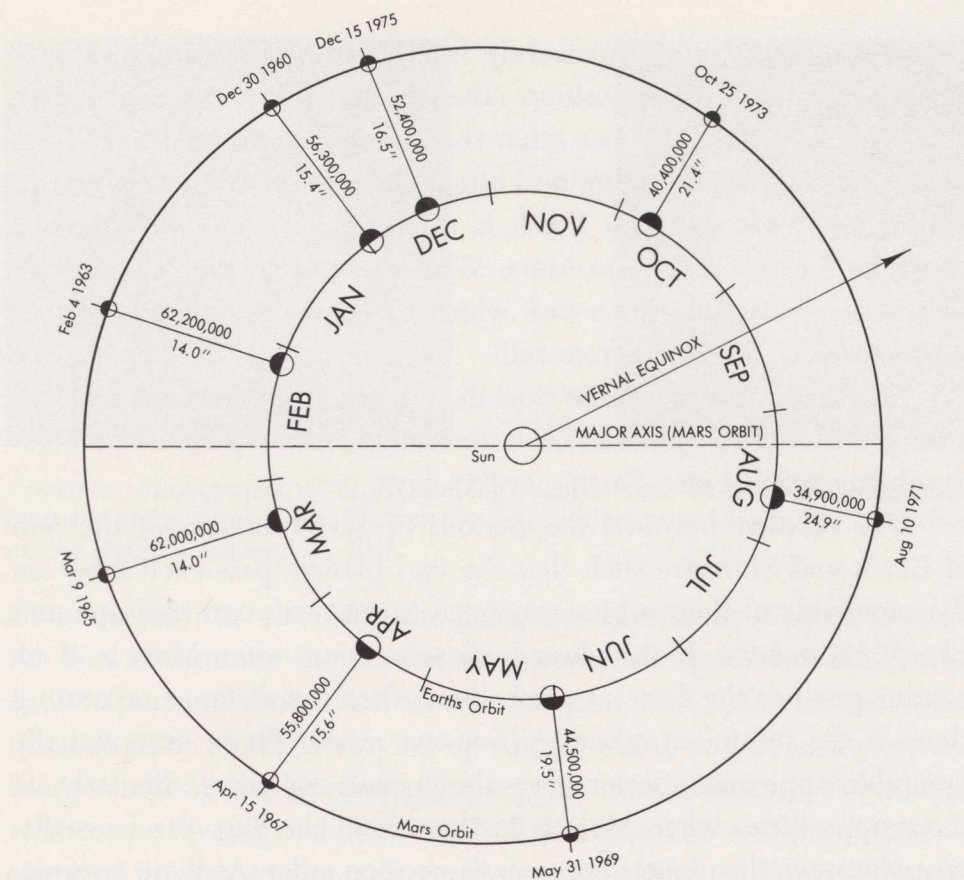


FIGURE 1. Diagram of the orbits of Mars and Earth. The next most favorable opposition will occur on August 10, 1971 when the two planets will be separated by 34,900,000 miles.

and Mercury; as a consequence, its distance from the Sun varies between 154,500,000 miles and 128,200,000 miles. This variation in solar distance is an important factor in explaining differences observed in the seasonal variation of the planet's snow caps and other phenomena. Its distance from Earth also varies greatly from 34,600,000 miles to 248,000,000 miles; this variation causes the apparent size of the planet's disk to range from 25".1 to 3".5 or a range of 50-fold in angular area. A disk with an angular size of 25 seconds of arc would appear as large as a dime (18 mm across) at 500 feet. The relative orbits of Mars and Earth are shown in Figure 1.

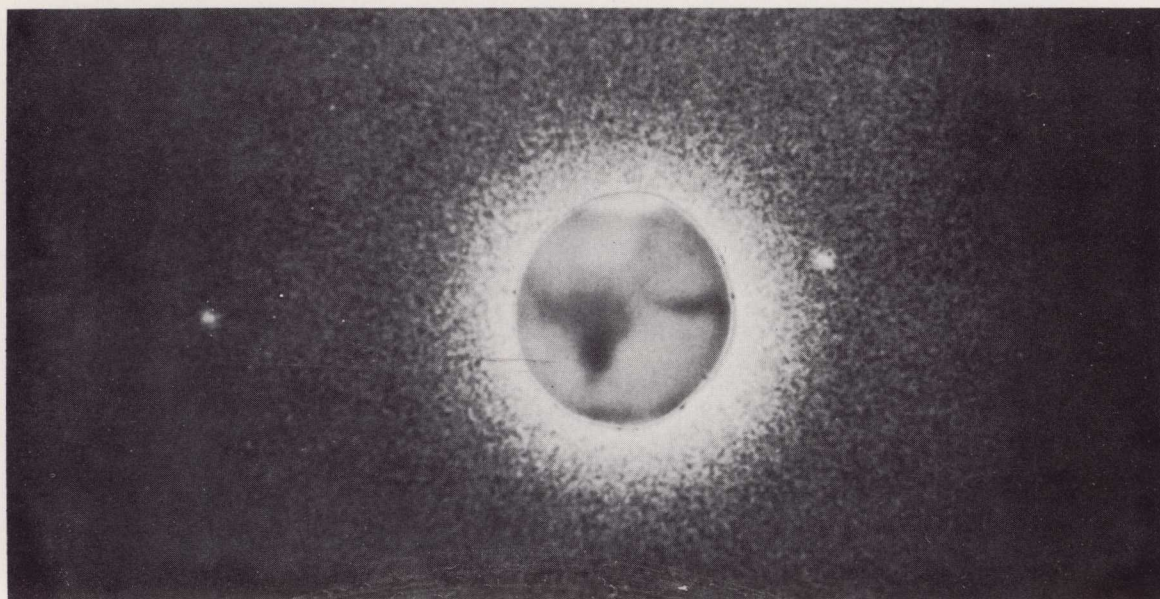


FIGURE 3. *Satellites of Mars photographed August 17, 1924. Phobos is close to the planet at the right. Deimos is considerably fainter and farther from the planet at the left. (Image of Mars printed separately.)*

are believed to be about 7 or 8 miles for Deimos, the outer one, and about 14 to 18 miles for Phobos, the inner one.

Their orbits, too, are the smallest of any bodies of the solar system and appear to be exactly circular with their planes in the Martian equator. The outer one (Deimos) revolves in a path 14,600 miles from the center of Mars in 30 hours and 18 minutes, and the inner one (Phobos), in a path only 5,800 miles from the center of the planet in 7 hours and 39 minutes. Thus, Phobos is the only known satellite that revolves in a period shorter than that of its planet. Its excessively short period causes Phobos to rise in the west, hasten across the Martian sky and set in the east after about $5\frac{1}{2}$ hours. Three times in the course of a Martian day Phobos rises and sets. Indeed Asaph Hall, when he first observed an inner satellite, at first thought that there were two or three inner moons. Deimos does not do this; it rises in the east like the stars, but its orbital motion eastward among the stars is so nearly equal to its westward motion, imparted by the



PHOTOGRAPH OF MARS obtained on August 24, 1956 by R. B. Leighton of the California Institute of Technology. The Mt. Wilson 60-inch reflector was used and its aperture was cut to 21 inches with an off-axis diaphragm. The exposure time was 20 seconds on Kodachrome Type A film. The positive, used in making the print, was composed by the Jet Propulsion Laboratory.

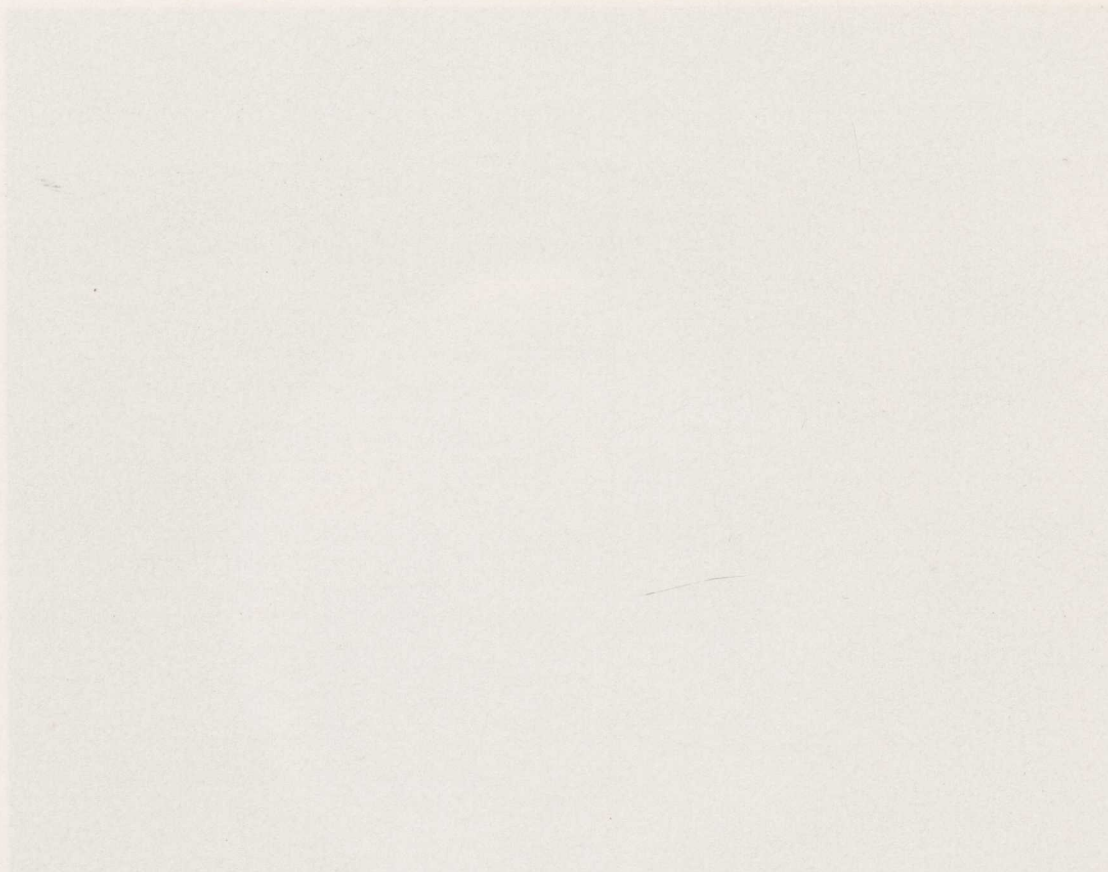


Plate I

This face of Mars is famous for the fact that Huygens first observed the Syrtis Major near the central meridian of Mars on November 28, 1659, at 7 hours in the evening; it is therefore the first marking ever recorded on the planet. This region is also especially notable for having displayed some of the most remarkable changes in the dark markings of any part of the planet.

In this and in all of the photographs of Mars appearing in this volume the direction north is toward the bottom and west is to the right.

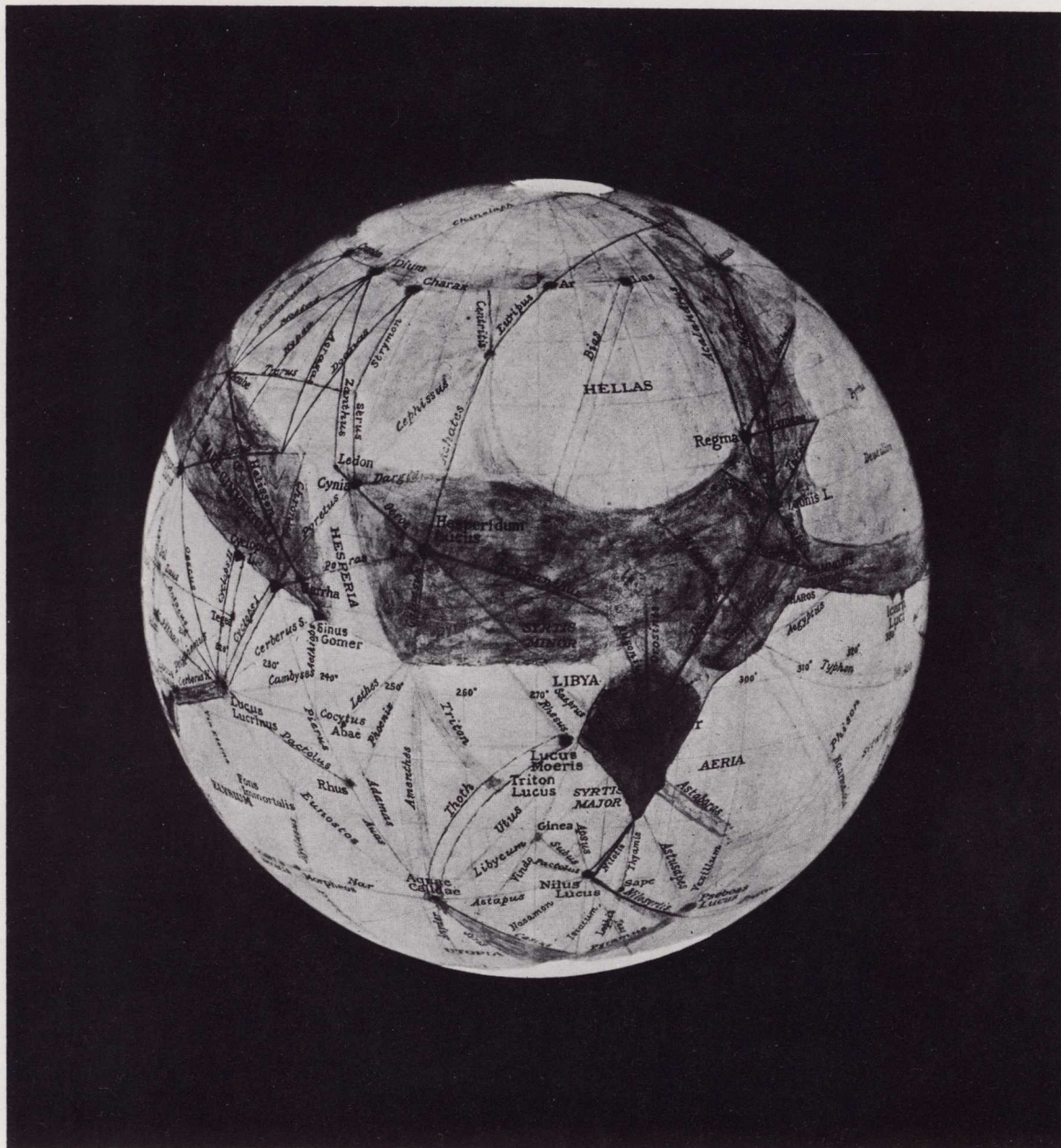
Noteworthy examples of change have been the vanishing, reappearance, and subsequent great expansion of the Thoth-Nepenthes system of canals, and the advent of a new dark area which developed to great prominence near Nodus Lacontis at the 1954 opposition. This region ^{1,2,3} has long been the scene of numerous frost and cloud formations, which, on occasions, have concealed the Syrtis Major from view, (see No. 3, Plate XXI); while further to the south the Hellas region has often been temporarily shrouded in white, first strikingly photographed at Lowell on August 1 and 2, 1909. Numerous examples of drifting clouds have been observed in the Isidis-Libya region.

¹Slipher, E. C., *Pop. Ast.*, 29, 2, 1921.

²_____, *P.A.S.P.*, 33, 193, 1921.

³Slipher, V. M., *Ibid*, 39, 147, 1927.

PLATE I
LOWELL'S GLOBE OF MARS 1907



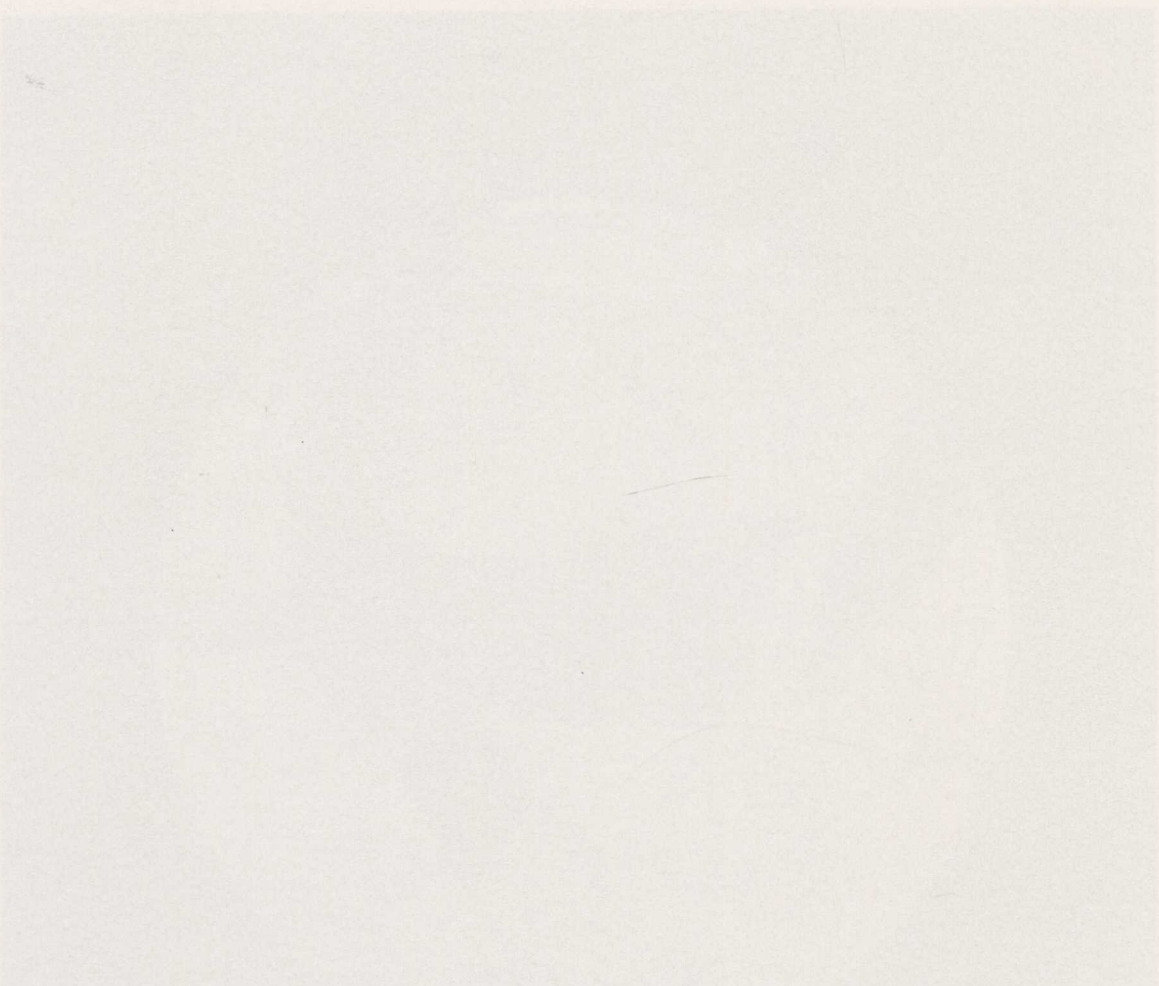


Plate II

The cameras used to obtain all Lowell photographs are equipped with specially figured negative lenses mounted in a movable carriage inside the tube. The internal mechanism is operated from the outside by dials for setting the lens for different magnifications and for focusing. Each camera is equipped with a double-slide plate carriage to enable the observer to make successive images quickly across the plate and to take several rows of images on a single plate. Mounts are provided for holding the color filter just in front of the negative lens. Ordinarily the camera is set to produce equivalent focal lengths of 160 to 220 feet. For blue, violet or ultra-violet photographs, it is necessary, of course, to employ a special blue-corrector lens ahead of the camera. The eyepiece attachment and camera can be interchanged easily in about one minute.

PLATE II
PLANETARY CAMERA ATTACHED TO THE 24-INCH LOWELL REFRACTOR

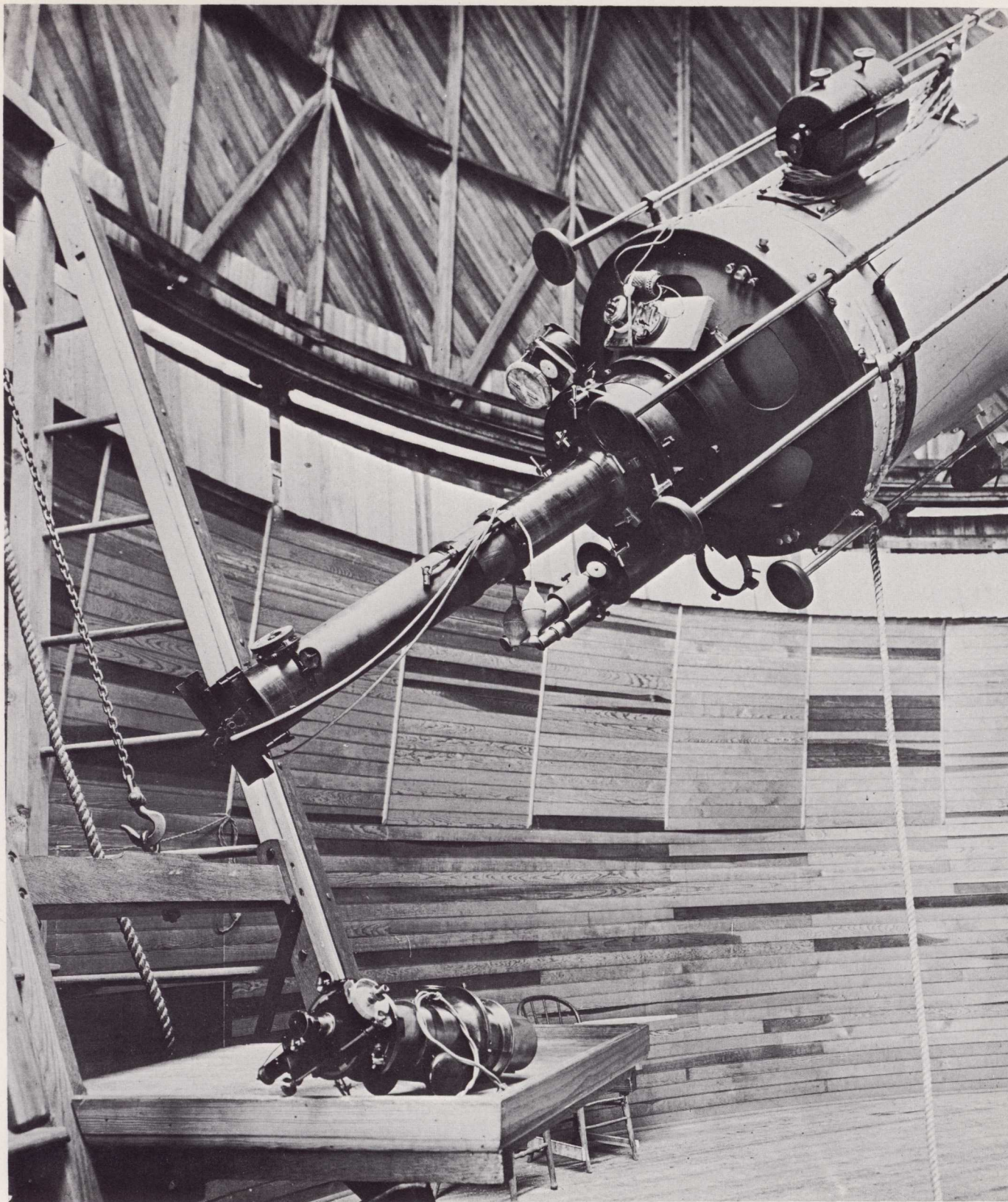




Plate III

This photograph, taken only a few days before opposition, December 4, 1911, shows the disk of Mars just after occultation by the moon. It reveals that, even at a better-than-average opposition, the planet's apparent disk is no larger in area than a medium-to-small crater on the moon. This serves to emphasize the meticulous care necessary to discover the hundreds of details that have been recorded on Mars and which confirm the numerous changes that have been observed.

PLATE III
OCCULTATION OF MARS BY THE MOON



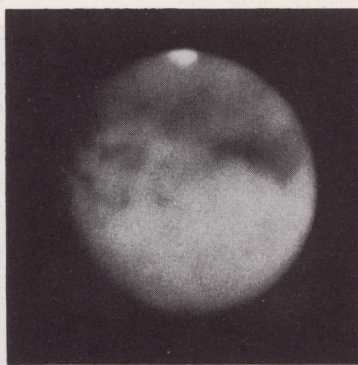
PLATE IV

REFERENCE MAP OF MARS AND ASSOCIATED PHOTOGRAPHS OF PRINCIPAL REGIONS

To facilitate the identification of markings referred to in the text, an annotated map of Mars by Lowell is here surmounted by a series of photographs covering a complete circuit of the planet.



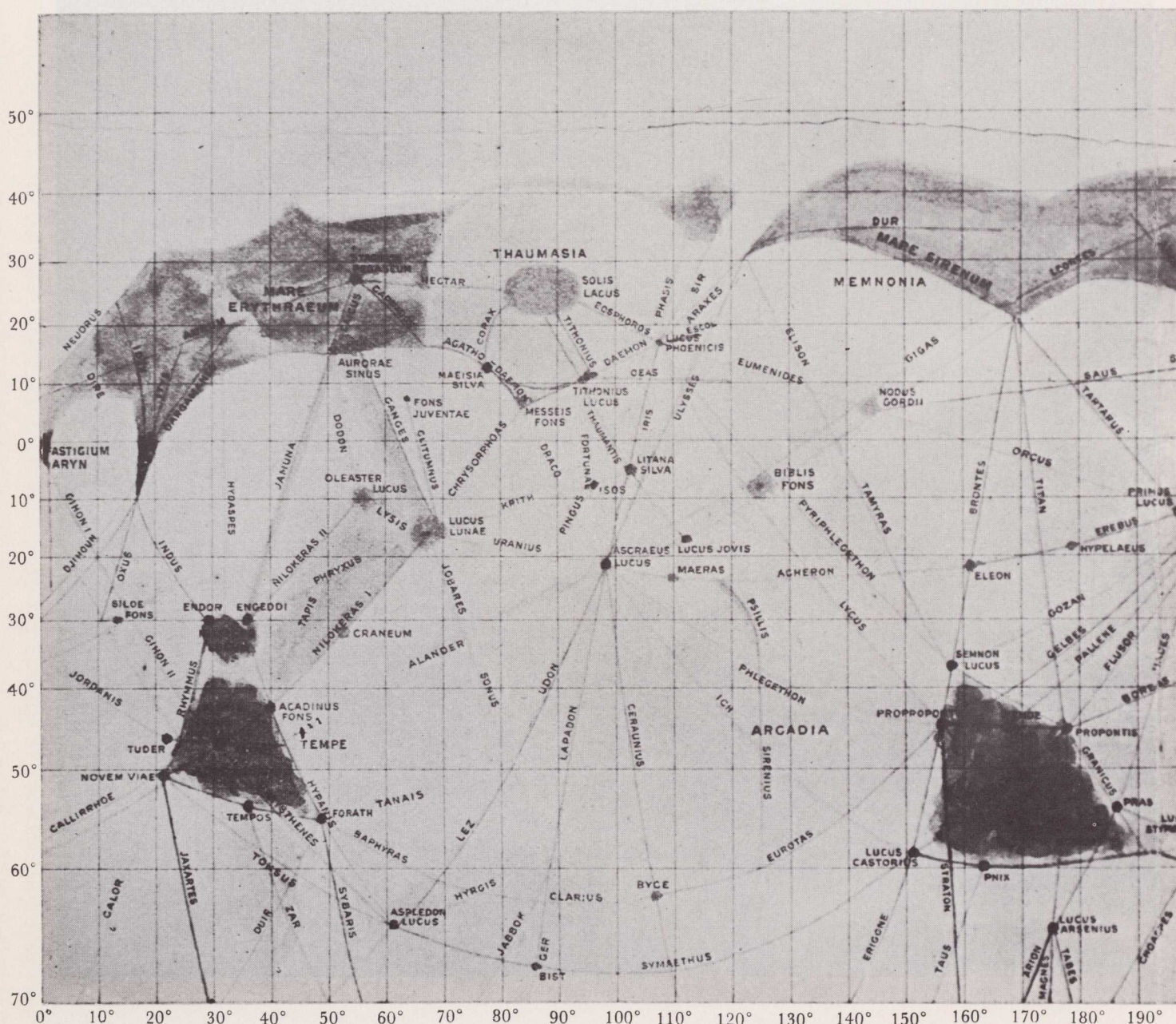
1. 1941 Oct 7
U.T. 10:45
July 9 M.D. $\lambda 48^{\circ}$
R



2. 1941 Sept. 25
U.T. 8:09
July 1 M.D. $\lambda 116^{\circ}$
O



3. 1941 Oct 23
U.T. 6:37
July 19 M.D. $\lambda 208^{\circ}$
Y



This annotated map of Mars, made by Lowell from visual observations in 1905, is intended to aid the reader in identifying certain markings referred to in the text. Since Mars presents to the Earth different tilts of its axis from one opposition to another, no one map

PLATE V

REFERENCE MAP OF MARS AND ASSOCIATED PHOTOGRAPHS OF PRINCIPAL REGIONS

Obviously the photographs and the map represent different oppositions, therefore, in making comparisons, allowance should be made for different tilts of the axis and other changes that occur with time.



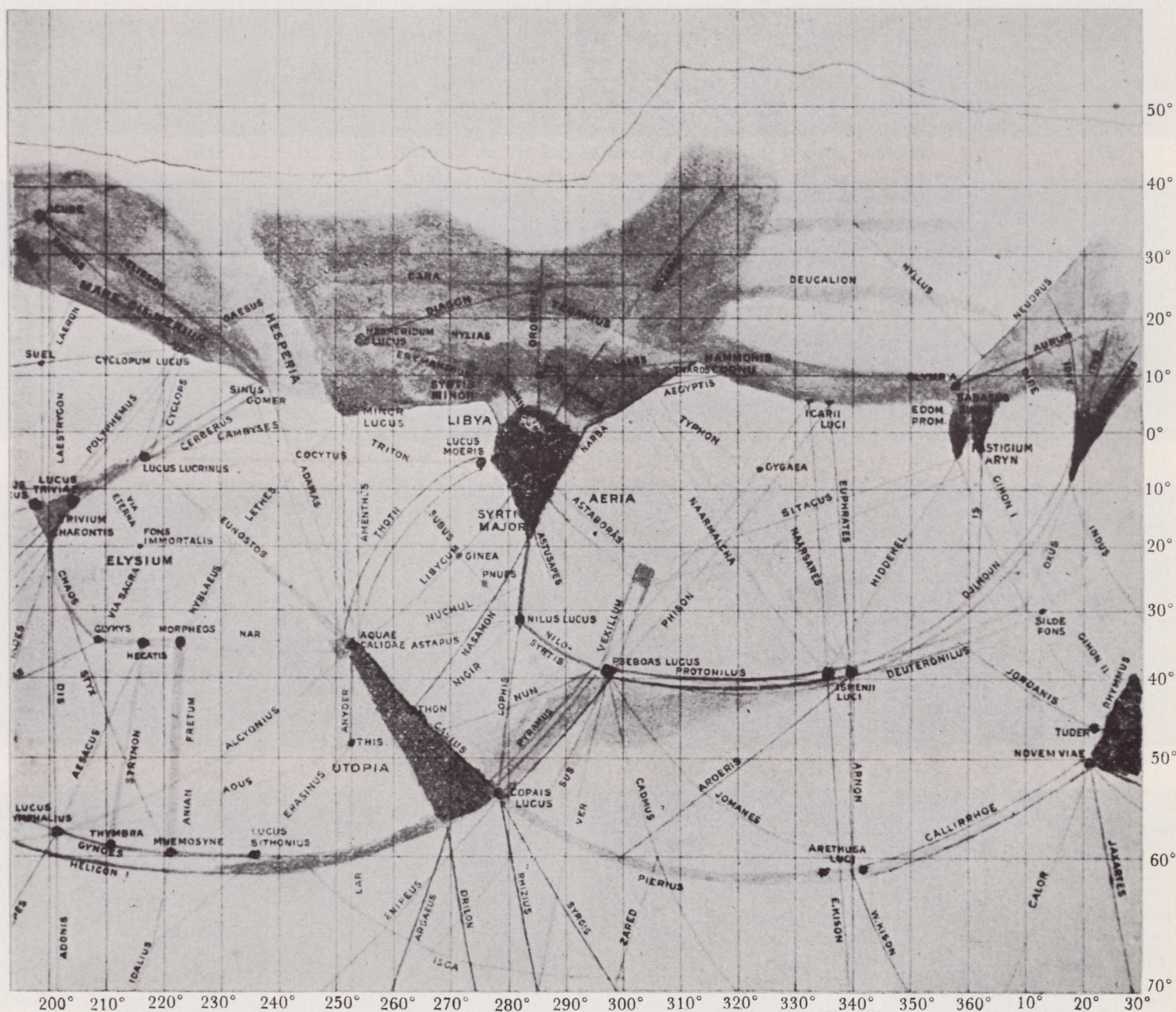
1. 1941 Oct. 17
U.T. 6:27
July 15 M.D. $\lambda 259^\circ$
Y



2. 1941 Oct. 17
U.T. 8:25
July 15 M.D. $\lambda 284^\circ$
R



3. 1941 Oct. 7
U.T. 5:16
July 9 M.D. $\lambda 328^\circ$
R



will suffice to show all its markings. Above the map is a series of 1941 photographs which corresponds to about every 60° of longitude on the map. These were made when the southern regions of Mars were well presented to view.

Plate VI

In 1907 Mars was more favorably placed for observation than it had been for more than a dozen years. The earth in its "stern chase" around the sun, caught up with Mars on July 12, 1907, and found it in the part of its orbit nearest the sun. The distance at opposition was 38,500,000 miles, not far from its smallest possible value. However, it was 25° south of the equator during the favorable time for observations; therefore it was unfavorably low in the southern sky for good observations from northern sites. Lowell, inspired by the initial success in photographing some of the canals at Flagstaff in 1905¹, and anxious to make full use of the next favorable opposition to continue the work, sent an expedition² to the Andes to observe Mars where it would culminate near the zenith each night. Fortunate in obtaining the loan of the excellent 18-inch refractor of Amherst College, the expedition set out from New York, by way of Panama on May 11 for an outdoor station at Alianza, Chile. The station was situated on the Tarapaca desert, elevation 4000 feet, in south latitude $20^{\circ}30'$, and inland about 70 miles from Iquique and beyond the coast range of the Andes. Here in this rainless and almost cloudless desert waste, the seeing was exceptionally fine.

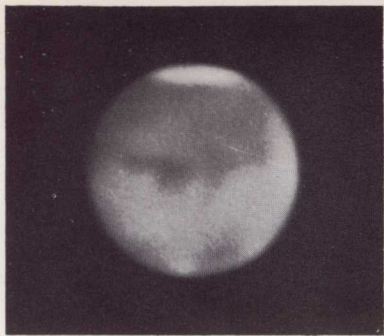
The accompanying photographs are representative examples from over 13,000 taken by the expedition. They have been prepared by optically superimposing about a half-dozen images from each plate into one composite picture, in order to minimize the effect of plate grain, but at the cost of losing much of the finer detail. Here the photographs are arranged in order of Martian longitudes of the disk, and cover one complete circuit of the planet.

The photographs show many interesting features on the disk and temporary atmospheric phenomena. Among these are: 1) temporary mists and clouds obscuring dark regions, (see veiled dark regions in Nos. 12 and 14); 2) night-to-night disappearance and reappearances of the canopy at the north pole demonstrating it consisted then chiefly of cloud; 3) the slow, gradual shrinking of the south cap with its fixed dark rifts and bright spots, revealing that it was a fixed deposit on the solid surface as contrasted with the cloud character of the north cap; 4) the temporary clouds in Southern Tharsis region near Lacus Phoenicis (these clouds have been much observed in later years especially in 1926 and 1954 when the configuration was referred to as the "W" cloud group); 5) morning limblight preceding Syrtis Major (see No. 20); and 6) definite traces of the majority of all canals shown in 108 complete drawings made from visual observations during the same period. On some of the best images, the original plates show as many as twenty-six of these delicate lines and oases. The Gehon, which was strong in 1907, and the Euphrates were shown to be double beyond doubt. This early series of photographs, supplemented by 4000 to 5000 secured at Flagstaff, have proved of great value in providing comparisons with later photographs. Note the great strength of Nilosyrtis in Nos. 17 and 18.

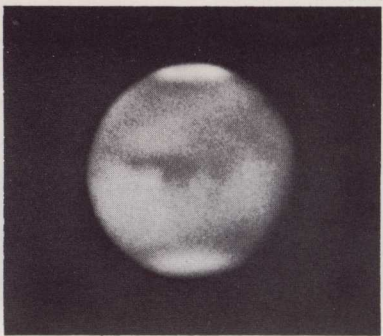
¹Lowell, P., Lowell Obs. Bull., I, 21, 1905.

²Members of the Expedition included Dr. David P. Todd, in charge; Mrs. Todd; A. G. Ilse, engineer of the Alvan Clark Corp.; R. G. Eaglesfield, electrician and general assistant; and the writer as observer.

PLATE VI
1907 CIRCUIT OF MARS



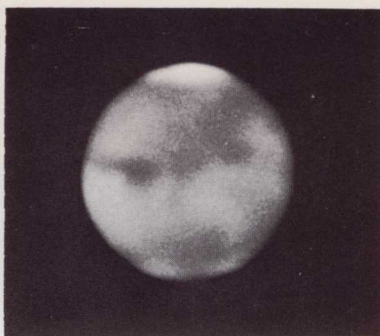
1. June 27 $\lambda 15^\circ$
U.T. 7:50 Apr 4 M.D. Y



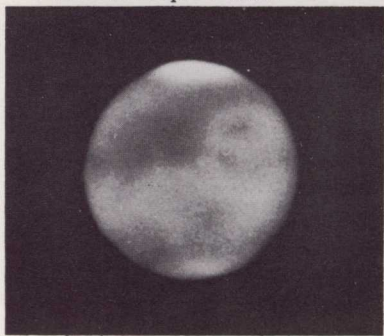
2. July 28 $\lambda 16^\circ$
U.T. 0:55 Y



3. July 27 $\lambda 22^\circ$
U.T. 1:57 Y



4. July 26 $\lambda 27^\circ$
U.T. 1:45 Y



5. July 24 $\lambda 56^\circ$
U.T. 2:35 Y



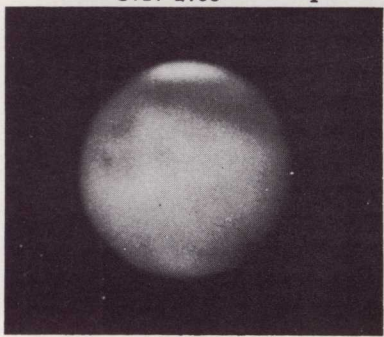
6. July 28 $\lambda 79^\circ$
U.T. 6:23 Y



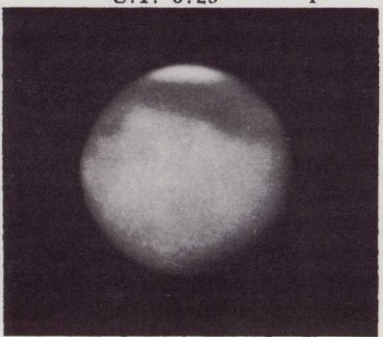
7. July 27 $\lambda 97^\circ$
U.T. 6:26 Y



8. July 26 $\lambda 120^\circ$
U.T. 8:13 Y



9. July 23 $\lambda 122^\circ$
U.T. 6:20 Y



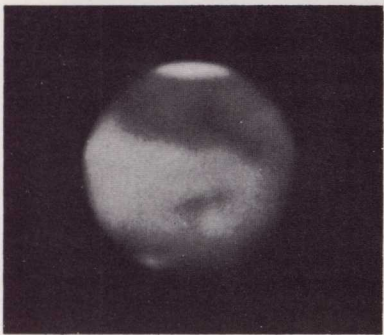
10. July 20 $\lambda 149^\circ$
U.T. 6:25 Y



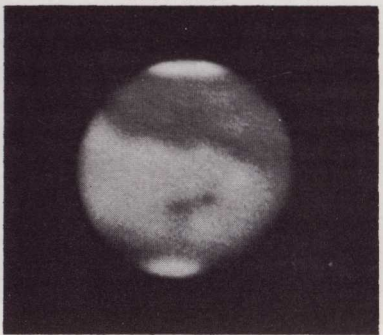
11. July 16 $\lambda 173^\circ$
U.T. 5:42 Y



12. July 17 $\lambda 196^\circ$
U.T. 7:45 Y



13. July 16 $\lambda 202^\circ$
U.T. 7:44 Y



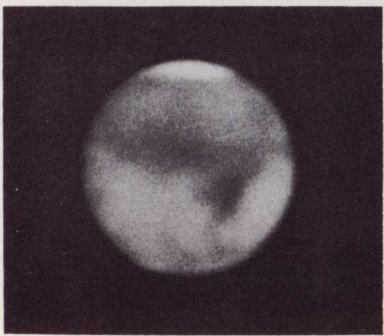
14. July 15 $\lambda 205^\circ$
U.T. 7:40 Y



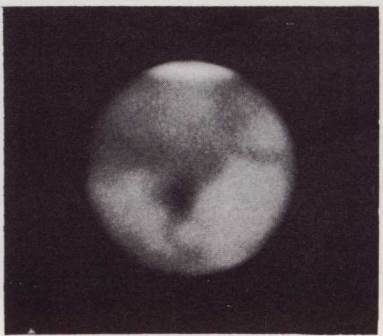
15. July 13 $\lambda 236^\circ$
U.T. 7:27 Y



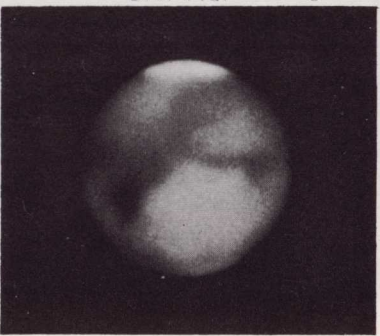
16. July 12 $\lambda 250^\circ$
U.T. 6:39 Y



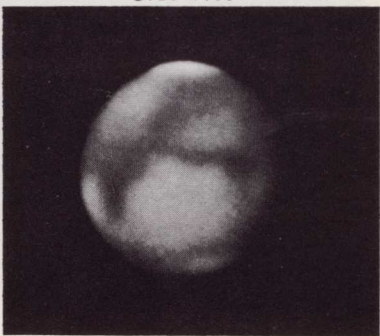
17. July 3 $\lambda 270^\circ$
U.T. 4:21 Y



18. July 3 $\lambda 294^\circ$
U.T. 5:56 Y



19. July 3 $\lambda 323^\circ$
U.T. 7:42 Y



20. July 30 $\lambda 331^\circ$
U.T. 0:56 Apr 23 M.D. Y

Plate VII

Following the initial efforts at Flagstaff to secure successful photographs of the planet in yellow light from 1903 to 1907, attempts to obtain photographs in other regions of the spectrum were begun in 1909 by making tri-color exposures in red, green and blue. The first row of images shows early examples of such tri-color photographs made in 1909. Results with the 42-inch reflector disclosed the surprising observations that blue images were completely blank, showed none of the permanent dark markings, and yet clearly enhanced the cloud cap and cloud areas. The green tri-color filter, which passed both green and blue light, (compensated filters were then not available) recorded the surface markings weakly as compared to the redder image which showed these in strongest contrast. Since similar tri-color photographs of paintings by Howard Russell Butler, N. A. gave identical results (second row), it was reasonable at that time to conclude that the explanation lay in the effect of selective reflection between the blue-green dark areas and the reddish deserts. This resulted in a complete leveling-out of the two in the blue photographs; the thought was that the thin atmosphere of Mars could have little or no effect.

The cause of the disappearance of the surface markings on the blue photographs, however, is less simple—there are at least two possibilities. One is that the actual colors of the reddish surface of the planet and its greenish-gray markings are such that in red light the first is much brighter than the second, while in blue light the red has lost much of its brightness and little contrast remains. The other is that the atmosphere of Mars, though transparent to red and fairly so to yellow, is very thick and hazy to blue light, and hence obscures the surface details.

No definite conclusion between these two alternatives seemed possible until 1937¹, when photographs by the writer revealed a state of blue clearing on May 20-23 so striking that the blue images revealed the surface details almost as clearly as ordinary yellow photographs. This example of blue clearing in the atmosphere of Mars led to the detection of others both before and since, with the result that the explanation of the "*violet layer*" and its temporary clearings has become a much discussed problem. A considerable number of explanations have been offered by physicists and physical chemists but none of these is free from serious objection. These are summarized elsewhere in this history.

The last row of images shows typical examples of modern tri-color photographs when the atmosphere of Mars was opaque to violet light. It is important to note here that there is no limb darkening in the blue photograph. Limb darkening is the fundamental requirement of the new theory announced recently by Opik² that carbon monoxide in the Martian atmosphere accounts for the violet layer.

¹Slipher, E. C., P.A.S.P., 49, 289, p. 137, 1937.

²Opik, E. J., Jnl. of Geo. Res., 65, 10, p. 3057, 1960.

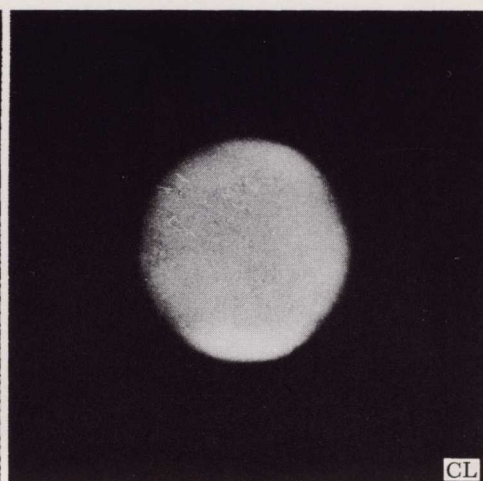
PLATE VII
EARLY TRI-COLOR EXPERIMENTS



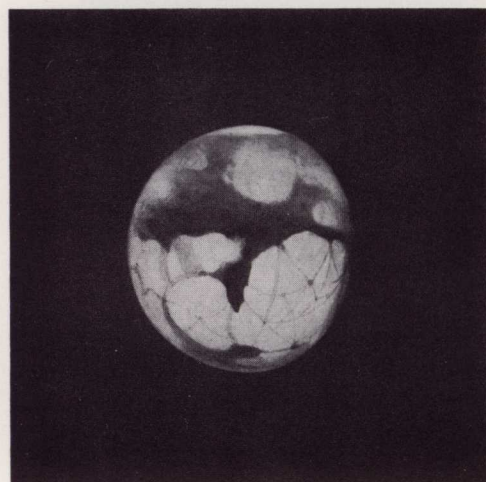
1. 1909 Sept 24 R



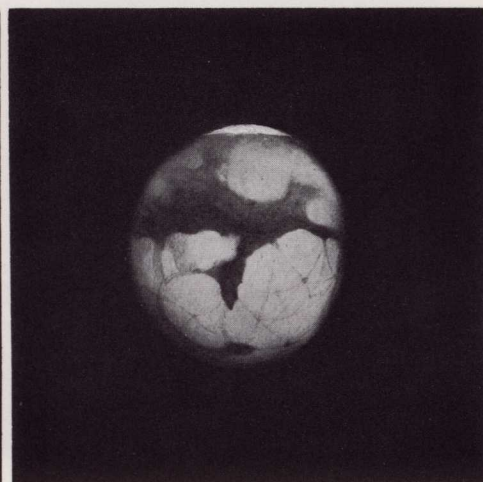
2. 1909 Sept 24 G-B



3. 1909 Sept 24 B



4. Butler Painting R



5. Butler Painting Y



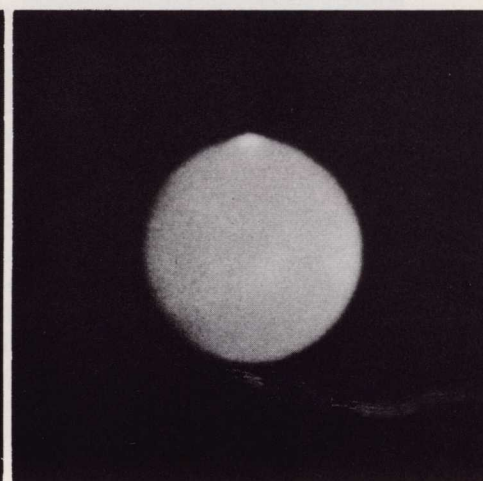
6. Butler Painting B



7. 1941 Oct 11 $\lambda 285^\circ$
U.T. 4:44 R

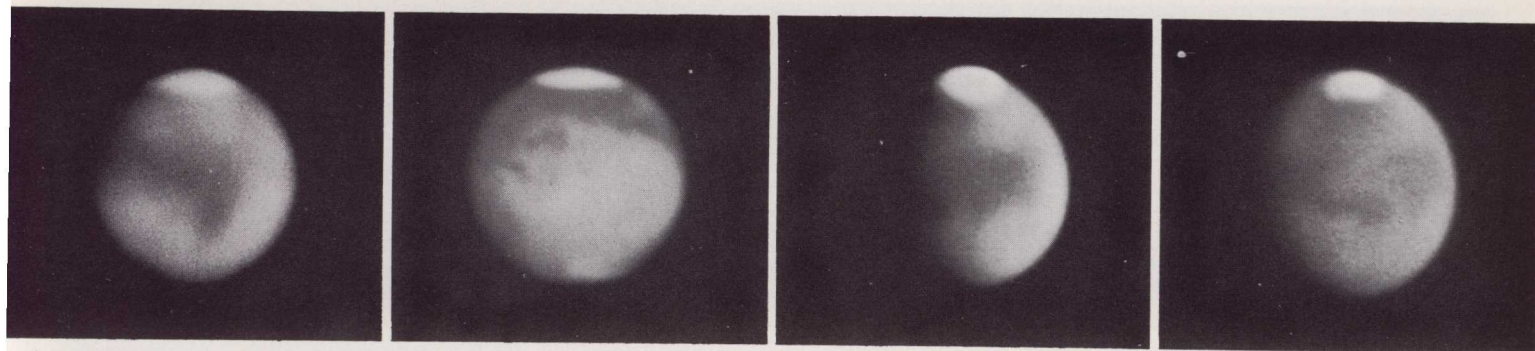


8. 1941 Oct 11 $\lambda 289^\circ$
U.T. 5:00 Y

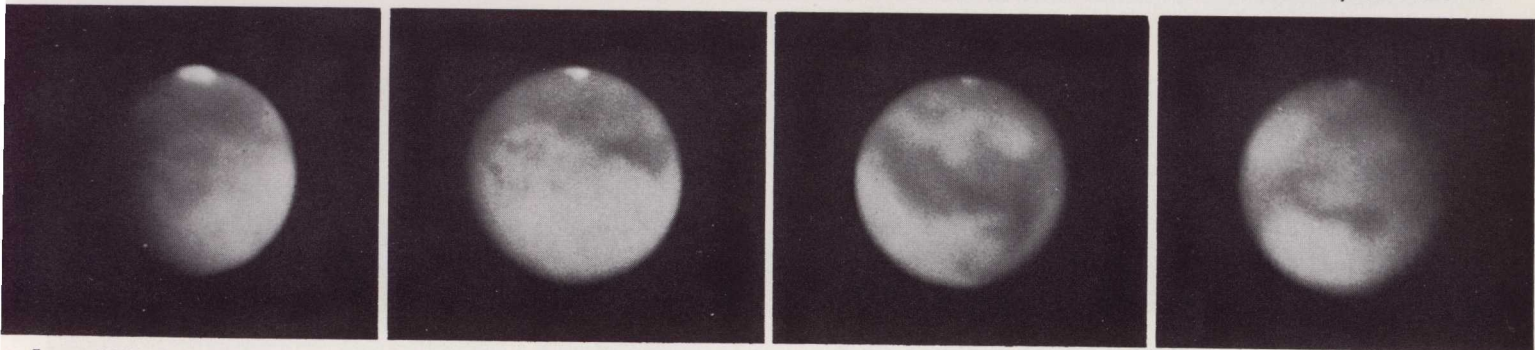


9. 1941 Sept 29 $\lambda 114^\circ$
U.T. 10:27 B

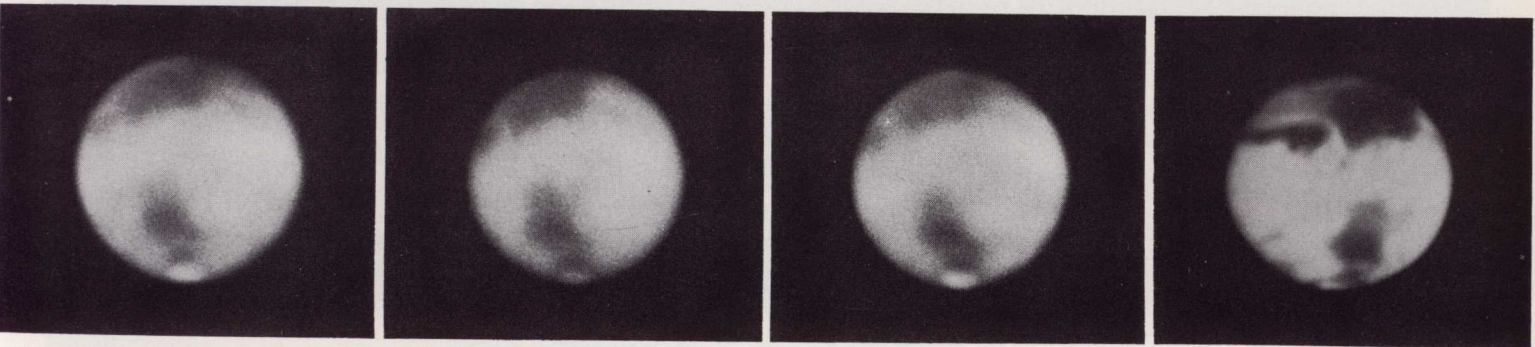
PLATE XII
SEASONAL MELTING OF THE SOUTH POLAR CAP, AND THE 'DARK COLLAR' WHICH ALWAYS HUGS
THE POLAR CAPS AS THEY RECEDE TOWARDS THE POLES DURING MARTIAN SUMMERS.



1. 1939 Apr 11 M.D. Y 2. 1939 Apr 21 M.D. Y 3. 1941 May 8 M.D. Y 4. 1941 May 26 M.D. Y



5. 1941 June 12 M.D. Y 6. 1941 July 1 M.D. O 7. 1941 July 15 M.D. Y 8. 1941 Aug 3 M.D. R



9. 1918 Mar 24 $\lambda 51^\circ$ Y
U.T. 8:20
Dec 22 M.D.
10. 1935 Apr 20 $\lambda 52^\circ$ Y
U.T. 4:06
Jan 15 M.D.
11. 1950 Mar 31 $\lambda 56^\circ$ R
U.T. 6:14
Dec 29 M.D.
12. 1952 May 22 $\lambda 15^\circ$ R
Feb 13 M.D.
200-in. Mt. Wilson-Palomar
Photograph

Plate XXXVII

These drawings made from visual observations may readily be compared with photographs of the same faces of the planet. The drawings were made at the Lick, Meudon and Lowell Observatories and the photographs were made at Lowell. Careful comparison of the five different sets of observations will disclose remarkable agreement between the best photographs and the drawings and also notable disagreements when the seeing was poor.

For the first pair (Nos. 1 and 2) Mars was near aphelion (apparent diameter $12''.95$) and the magnification of the image was equivalent to 1800 diameters. Despite the severe loss of detail resulting from image composition to reduce graininess, enough remains to verify the existence of most of the details in the drawings. Careful study of the original plate shows virtually everything seen visually.

Drawing No. 3 by Trumpler using the Lick 36-inch refractor on October 23 is substantially confirmed by my photograph, No. 4, of the same general region on November 22. Close comparison discloses that traces of most of the canals and oases on the drawing are corroborated by traces of similar detail in the photograph.

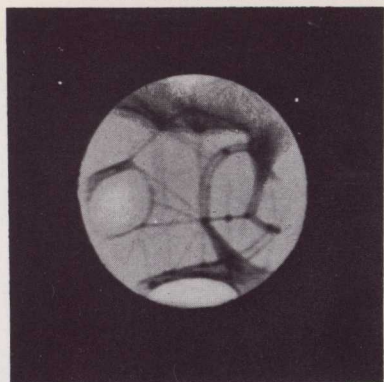
Comparison of the Antoniadi drawing, No. 5, with my photograph of the same region, No. 6, (magnification 1700 diameters) reveals that all the numerous details in the photograph to the left of and above the Thoth, namely in the Aethiopis and Elysium regions, are not recorded in his drawing. However, his representation of the aspect of the Thoth itself is not supported by the photograph. Examples are the short double canal running downward from the base of the Thoth, and also the dark, core-like medial line through the center of the Thoth flanked by diffuse irregular shadings on each side.

No. 7, is a circular section copied from Trumpler's 1924 map of the planet which is compared with our 1941 photograph, No. 8. A careful comparison should disclose that many of the finer details of his map are identifiable in the photograph, in particular many of the details in the southern part of the disk above the Sabaeus Sinus. The many fine traceries and spots in his map are faintly revealed in the original photograph.

No. 9 is a drawing by the author. Photograph No. 10 was made a little later and shows the same face of the planet as Nos. 5 and 6. Here the agreement between drawing and photograph is more satisfactory and the one corroborates the other in a substantial manner. The markings missing in Antoniadi's drawing should be apparent in our drawing and photograph. Astronomers who observe the planet both visually and photographically know that nearly every detail observed visually can be found on some of the photographs, but the latter may show only a trace of something without clearly revealing its character or appearance. The visual observations have been of great help in the representation of the fine structure, such as sharpness, width and regularity of boundary lines or canals. On the other hand many markings are quite satisfactorily shown by the photographs which were not noticed in the visual observations. Those experienced in both types of observations fully realize the photograph may not show the precise character of extremely minute markings, but with regard to the reality of controversial markings, the evidence of the photograph is undeniable.

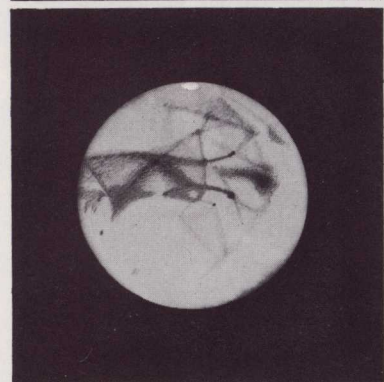
PLATE XXXVII
COMPARISON OF DRAWINGS AND PHOTOGRAPHS

1. 1916 Feb 11
(Slipher drawing)
Nov 15 M.D.



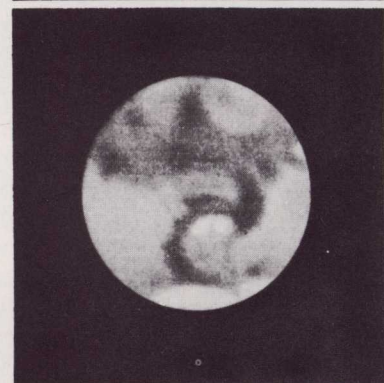
2. 1916 Feb 12 $\lambda 250^\circ$
U.T. 4:26 Y
Nov 15 M.D.

3. 1926 Oct 23
(Trumpler drawing)
July 30 M.D.



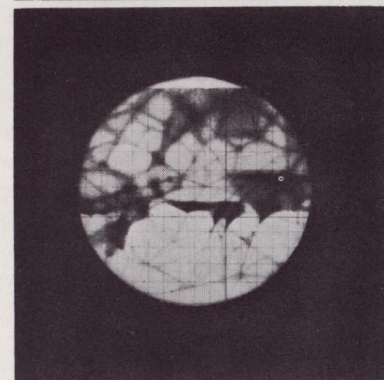
4. 1926 Nov 22 $\lambda 96^\circ$
U.T. 4:12 Y
Aug 16 M.D.

5. 1928 Dec 15
(Antoniadi drawing)
Sept 21 M.D.



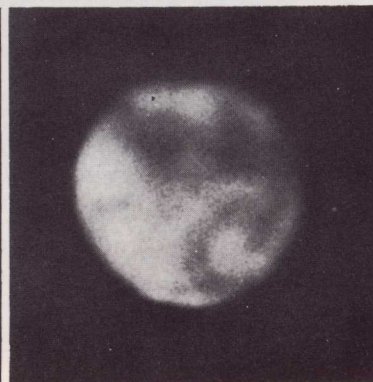
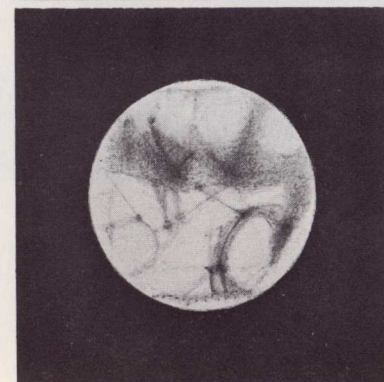
6. 1928 Dec 29 $\lambda 245^\circ$
Sept 28 M.D. Y

7. 1924 Trumpler Map

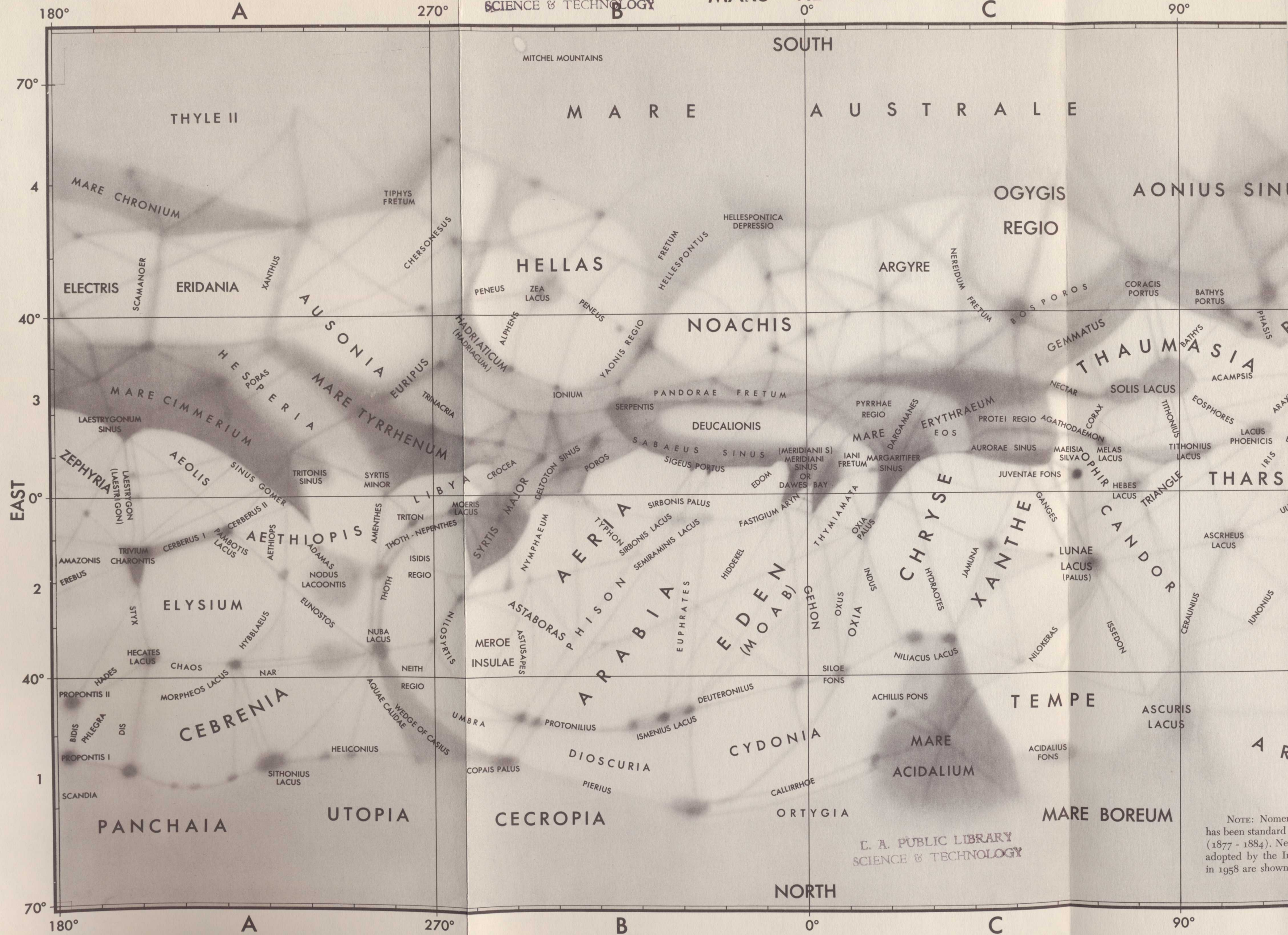


8. 1941 Oct 11 $\lambda 338^\circ$
U.T. 8:06 R
July 11 M.D.

9. 1928 Dec 28
(Slipher drawing)
Sept 28 M.D.



10. 1928 Dec 29 $\lambda 245^\circ$
Sept 28 M.D. Y



NOTE: Nomenclature has been standardized (1877 - 1884). Names adopted by the IUPAC in 1958 are shown.

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